

Section 1A. Course Introduction

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Academic background:

- ▶ Ph.D. MIT EECS

Current affiliations at the University of Pennsylvania:

- ▶ Department of Electrical and Systems Engineering
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Research Interest:

- ▶ Network analytics
- ▶ Data science
- ▶ Stochastic modeling
- ▶ Control of dynamical systems



Data Science is about drawing useful conclusions from large and diverse data sets through *exploration*, *prediction*, and *inference*:

1. **Exploration** involves identifying patterns using visualization and descriptive statistics
2. **Prediction** involves using information to make informed guesses about values we wish we knew using machine learning
3. **Inference** involves quantifying our degree of certainty: How accurate are our predictions?
The main tools for inference are statistical tests and models

Statistics is a central component in Data Science because it studies how to make robust conclusions based on incomplete information. **This is the focus of this course**

Two different approaches to explain reality:

- ▶ ***Model-based approach***: A classical approach based on deriving models from first principles, such as Newton's law, relativity, optics, etc.
- ▶ ***Data-driven approach***: In many modern problems, it is not possible to explain the behavior of a system from first principles, e.g. stock market, human behavior, the brain, etc. In this situation, we can still make useful predictions using data science

In practice, the best approach is to combine both models and data

Let us look into a **few examples...**

Examples of Data-Driven Learning Problems

A few practical examples of statistical learning:

1. Identify the numbers in a handwritten zip code
2. Establish the relationship between salary and demographic variables
3. Classify recorded phoneme
4. Classify land usage in LANDSAT image
5. Risk factors in heart disease
6. Understanding COVID-19
7. Others: Stock market, machine failure, etc.

Can you design an algorithm able to recognize the numbers in these gray-scale low-resolution images?

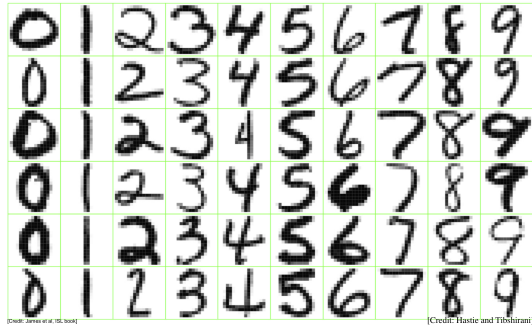


Figure: 16x16 grey-scale images of handwritten letters.

Given a dataset containing the salaries of individuals, as well as their age, year, and educational level, can you design an algorithm able to predict the salary of an individual (not in the dataset)?

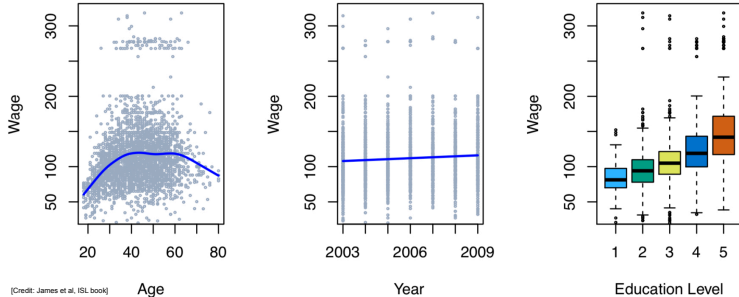


Figure: Income survey data for males in the USA, 2009.

Can you design an algorithm to recognize a particular phoneme from a sound signal (or its power spectral density)?

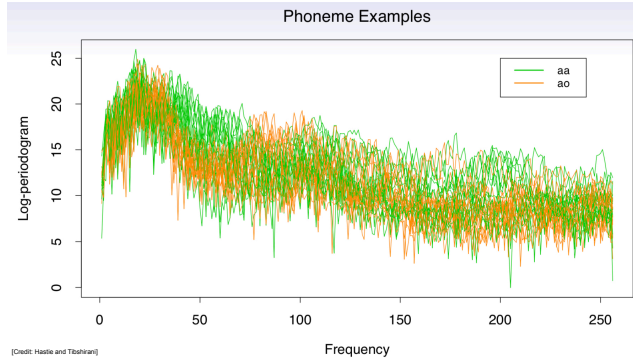


Figure: Power spectral densities for sounds “odd” (green) and “ought” (orange)

Can you design an algorithm able to recognize land usage (red soil, gray soil, vegetation, cotton, etc.) from an image taken using four different spectral bands?

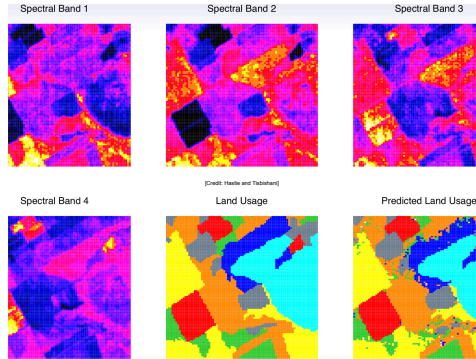


Figure: Satellite images at 4 spectral bands; actual and predicted land usage.

Given a dataset containing a collection with and without heart disease, as well as their obesity level, alcohol consumption, and age, can you estimate the probability that an individual (not in the dataset) develops heart disease by measuring these variables?

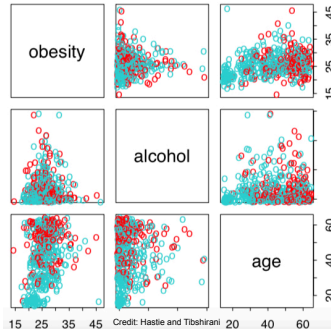


Figure: Scatter plot matrix for heart disease dataset.

Given community mobility reports from Google Maps and a time series of the number of infections/deaths/hospitalized due to COVID-19, can you explain how mobility patterns affect the dynamics of the disease?

Retail & recreation

-24%

compared to baseline



Grocery & pharmacy

-1%

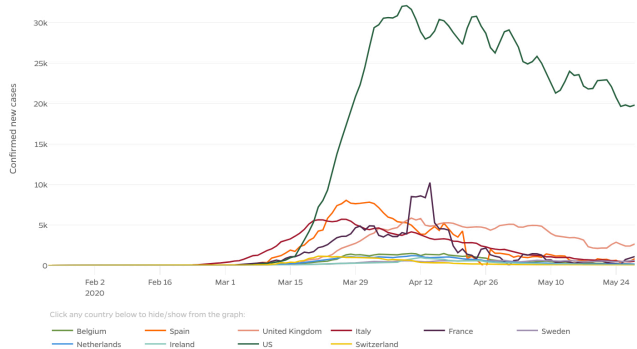
compared to baseline

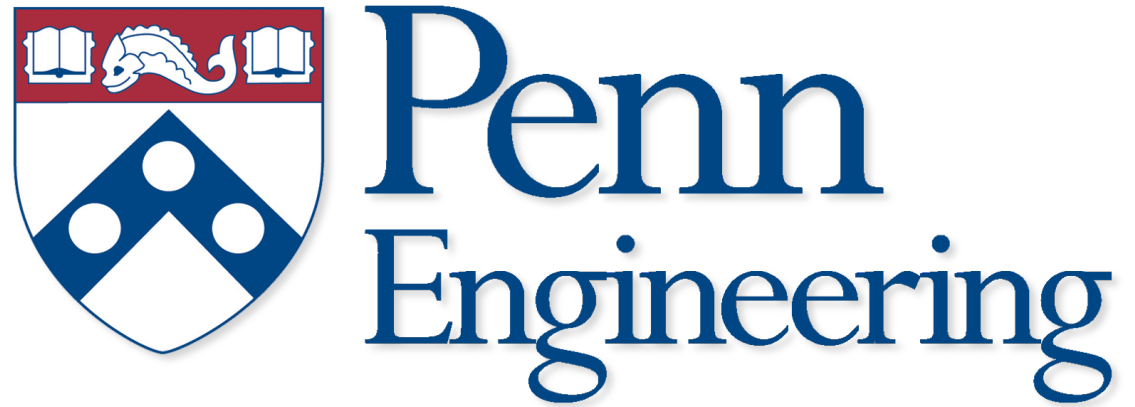


Parks

+88%

compared to baseline





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